PPP-C-55a
April 16, 1975
SUPERSEDING
Fed Spec. PPP C-55
October 16, 1959

FEDERAL SPECIFICATION

CANS, COMPOSITE, FOR DRY PRODUCTS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for composite cans, with ends, suitable for use as unit packages in the shipment and storage of dry supply items.

1.2 Classification.

1.2.1 Types, classes, shapes and styles. The cans covered by this specification shall be of the following classes, types shapes and styles as specified (see 6.2).

Type I - Spiral Wound
Type II - Convolute Wound
Type III - Lap Seam

Class 1 - Food product container.
Class 2 - Non-food product container.

Shape A - Round
Shape B - Square
Shape C - Rectangular
Shape D - Oval
Shape E - Oblong

Style 1 - Plain end
Style 2 - Easy open end
Style 3 - Single or multiple friction plug end
Style 4 - Cap or slip cover
Style 5 - Turn or sifter top
Style 6 - Screw top
Style 7 - Curled and disc end

1.2.2 Sizes. Size of containers shall be as specified in the contract or order and, unless required for a special purpose shall be a commercial size as regularly supplied by the industry (see 6.2 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications

QQ-T-425 - Tinplate (Hot Dip and Electrolytic)
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard.

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Federal Standard

Fed Std. No. 123 - Marking for Domestic Shipment (Civil Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification

MIL-L-10547 - Liners, Case - Waterproof.

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations

21 CFR 121 - Federal Food, Drug and Cosmetic Act and Regulations promulgated thereunder.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issurance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Incorporated, Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P. Street, N.W., Washington, DC 20036.)

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Composite Can and Tube Institute

CT-101 - Measuring Wall Thickness
C-110 - End Blow-Off Pressure

(Application for copies should be addressed to the Composite Can and Tube Institute, 1725 Eye Street, N.W. Washington, DC 20006.)

3. REQUIREMENTS

3.1 Material. The material used for constructing containers furnished under this specification shall conform to the applicable requirements specified herein, and shall be free from any defects which may impair serviceability or proper functioning under service conditions, or which may result in contamination which renders the product unfit for use. Materials used in the construction of class 1 containers shall comply with the Federal Food, Drug and Cosmetic Act, and Regulations promulgated thereunder.

3.1.1 Adhesives. Adhesives used shall not contaminate the packaged product and shall permit performance in accordance with the requirements listed below.

3.1.2 Structural plies. Structural plies shall be of fiberboard or paperboard of sufficient thickness and strength to enable the cans to comply with the requirements of Table I.

3.1.3 Ends. Ends shall be of paper or paperboard, tinplate conforming to QQ-T-425, type II, grade 4, class 25, black plate, tin free steel, aluminum, or plastic (thermoplastic or thermosetting).

3.1.4 Inner barriers. Inner barriers, where necessary for product protection and containment, shall be combinations of paper and metal foil, combinations of paper and plastic film, or treated papers. Unless otherwise specified (see 6.2), the use of and type of inner barrier shall be at the option of the manufacturer.

3.1.5 Labels. Labels shall be combinations of paper and metal foil, combinations of paper and plastic film, or treated paper which will withstand normal handling and provide container protection and product identification.

3.2 Fabrication.

3.2.1 Body. The can body shall be either spirally or convolutely wound or of lap seam construction, and constructed of materials specified under 3.1.

3.2.2 Spiral cans. Body seams of spiral wound cans shall be overlapped, slived and overlapped, or butted. Where necessary, butt joint gaps shall be filled with a suitable flexible material.

3.2.3 Inner barrier overlap. Inner barrier edges shall be folded under, lapped, or otherwise sealed when necessary for product containment and protection.

3.2.4 Adhesive bonds. All adhesive bonds shall be 80 percent minimum fiber tearing bonds when tested in accordance with 4.6.1.

3.2.5 Wall thickness and strength. The body wall shall be constructed of materials conforming to the minimum body wall thickness requirements of Table I when tested in accordance with 4.6.2.

3.2.6 Size. The can size shall be as specified (see 6.2) and shall be a commercial size as regularly supplied by industry.

3.3 Ends.

3.3.1 Style 1, Plain end. Style 1 ends shall be made of metal as specified in 3.1.3 and shall conform to the requirements of Table I.

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TABLE I. Requirements for Types I, II, and III Cans

Cross Sectional Area of Body (square inches)	Outside Diameter of Round Cans	Minimum Body Wall Thickness (inches)	Minimum Average Base Weight of Steel End Plate or Coil (lbs. per base box)	Minimum Average Thickness of Aluminum End Sheet or Coil (inches)	Minimum Average End Blow-off 1/ (lbs. per sq. inch)
Up to 5	Up to 208	0.015	55	0.0075	15
5 to 7.5	209 to 301	0.020	55	0.0075	13
7.5 to 12.5	302 to 315	0.025	60	0.0080	12
12.5 to 20	406 to 500	0.030	60	0.0090	10
20 to 29	501 to 600	0.035	70	0.0100	7
29 to 40	601 to 702	0.035	90	0.0120	4

1/ Applies only to style 1 ends.

3.3.2 Style 2 - Easy open end. Style 2 ends shall be made of metal as specified in 3.1.3 and shall not conform to the end blowoff requirements of Table I. The easy open end shall be fabricated in any manner which permits the can to be opened by hand and allows the contents to be easily removed. Easy open ends shall not be reclosable. Each can shall be marked with instructions for proper opening and disposal of the end and warnings against the danger of sharp edges.

3.3.3 Style 3 - Single or multiple friction plug end. Style 3 ends shall be made of metal or plastic as specified in 3.1.3 and shall not conform to the end blowoff requirements of Table I. Closure shall be accomplished by single or multiple friction and shall be of proper, tight fit.

3.3.4 Style 4 - Cap or slip cover end. Style 4 ends shall be made of paper, plastic, or metal as specified in 3.1.3 and shall not conform to the end blowoff requirements of Table I. The cap or cover shall be drawn and provided with a skirt having a minimum length of 1/2 inch that can be slipped over the open end of the can. Fit shall be sufficiently tight so that the cap or cover will not fall off by its own weight when the empty can is held in the inverted position.

3.3.5 Style 5 - Turn or sifter top end. Style 5 ends shall be made of metal, plastic, or a combination of metal and plastic as specified in 3.1.3 and shall not conform to the end blowoff requirements of Table I. The turn or sifter top shall be a commercially available size with the number of openings at the option of the manufacturer.

3.3.6 Style 6 - Screw top end. Style 6 ends shall be made of metal as specified in 3.1.3 and shall not conform to the requirements of Table I. The cover shall be threaded, and the open end of the can shall be threaded or provided with a threaded metal sleeve, at the option of the manufacturer, to receive the cover and provide a tight fit.

3.3.7 Style 7 - Curled and disc end. Style 7 ends shall be made of paper as specified in 3.1.3 and shall not conform to the end blowoff requirements of Table I. The open end of the can shall be flanged and the curled and disc end shall be formed in such a manner that it will tightly secure itself to the flange.

3.4 Secondary closure. When specified (6.2), the can shall be provided with a secondary closure made of paper or plastic as specified in 3.1.3. The secondary closure shall not be bonded to the body walls in order to be used as the can closure after a non-reclosable end has been opened and removed. The secondary closure shall provide a sufficiently tight fit so that it will not fall off by its own weight when the empty can is held in the inverted position.

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3.5 Marking of cans. When specified (see 6.2), the cans shall be marked as required by the procuring agency. When applicable, the cans shall be marked in accordance with the Federal Food, Drug and Cosmetic Act and Regulations promulgated thereunder.

3.6 Workmanship. All finished cans shall be clean, free of foreign matter, and free from any defects which may affect the appearance or serviceability, such as surface scuffing, inadequate adhesion between plies, ragged joints, recutter cuts, imperfect seams or other defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection lot. Unless otherwise specified, a lot shall consist of all cans of the same types and sizes from one manufacturer, offered for delivery at one time.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with MIL-STD-105, except as otherwise indicated.

4.4 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, standards and regulations unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.5 Inspection of cans. Cans selected in accordance with 4.3 shall be inspected for the following defects. The sample unit shall be one can. The inspection level shall be S-2 with an acceptable quality level (AQL) of 2.5 expressed in terms of defects per hundred units for major defects and an AQL of 6.5 for minor defects.

Examine		Category Major	Minor
Class	Not as specified.	x	
Type	Not as specified.	x	
Size	Not as specified	x	
Style	Not as specified	x	
Workmanship	Not in conformance with 3.4.		x
Marking	Not as specified		x

4.6 Testing. Cans selected in accordance with 4.3 shall be tested in accordance with the following subparagraphs. The sample unit shall be one can, and the inspection level which determines sample size shall be S-3. Test requirements apply to lot averages.

4.6.1 Adhesive bonding. Using a sharp knife, cut the ends off the can body as close to the end seams as possible. Split the can body lengthwise so that it can be flattened out as a rectangle. Peel apart the liner overlap, liner, body plies, label and label overwrap. Measure any area where there is no fiber tearing adhesive bond and calculate the percentage of the total area involved to determine compliance with 3.2.4.

4.6.2 Body wall thickness. The can body wall thickness shall be measured using Composite Can and Tube Institute (CCTI) method CT-101, to determine compliance with 3.2.5 and Table I.

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4.6.3 End plate. The thickness of the can end plate shall be measured with a suitable micrometer caliper to determine compliance with Table I. The weight of the steel end plate may be calculated from its thickness using Table IV of QQ-T 425.

4.6.4 End seam blowoff. The can ends shall be tested to determine conformance with the end blowoff requirements of Table I, using CCTI method C-110.

4.7 Examination of preparation for delivery. An examination shall be made to determine that the packing and marking comply with the requirements of Section 5 of this specification. Defects shall be scored in accordance with Table II. For examination of packing, the sample unit shall be one shipping container fully prepared for delivery, just prior to the closing operation. Defects of closure listed in Table II shall be examined on shipping containers fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

TABLE II. Classification of defects

Examine	Defect
Markings	Omitted incorrect, illegible improper size location, sequence or method of application.
Materials	Any nonconforming component component missing, damaged or otherwise defective affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flaps inadequate nailing and stapling, loose strapping, or distortion of container.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2)

5.1.1 Level A. Cans shall be packed in a close-fitting box conforming to PPP-B-585, class 3 PPP-B-591, class 2, PPP-B-701, overseas type or PPP-B-621 class 2. Separators made of fiberboard conforming to PPP-F-320, class weather resistant, type SF or CF (variety SW), grade W5 or W6 shall be used between tiers of cans. Each shipping container shall be provided with a type I or IT, grade B, class 2 caseliner conforming to MIL-L-10547, closed and sealed in accordance with the appendix to the applicable box specification.

5.1.2 Level B. Cans shall be packed in a close fitting box conforming to PPP-B-636, class domestic, type CF, variety SW, grade 275 minimum, style optional. Fiberboard separators conforming to PPP-F-320, or paperboard separators shall be used between tiers of cans. The box shall be closed in accordance with the appendix to the box specification.

5.1.3 Level C. Cans shall be packed in a manner that will assure acceptance and safe delivery to destination in compliance with the Uniform Freight Classification rules or the National Motor Freight Classification rules, as applicable.

5.2 Marking. Marking shall be in accordance with 5.2.1 or 5.2.2 as specified (see 6.2).

5.2.1 Civil agencies. In addition to any special marking required by the contract or order each shipping container shall be marked in accordance with Fed Std No 123.

5.2.2 Military activities. In addition to any special marking required by the contract or order, each shipping container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Cans conforming to this specification are suitable for use as unit packages for the shipment and storage of dry supply items.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

(a) Title, number and date of this specification.
(b) Type, class, shape, style and size of cans (see 1.2.1, 1.2.2 and 3.2.6).
(c) When a specific inner barrier is required (see 3.1.4).
(d) When a secondary closure is required (see 3.4).
(e) Marking on cans, when required (see 3.5).
(f) Level of packing (see 5.1).
(g) Special shipping container marking, when required (see 5.2.1 or 5.2.2).

6.3 Sizes. When specifying sizes, the can manufacturing industry uses a three digit number. The first digit shows the integer of the closest, smallest whole number of the dimension and the second and third digits show the additional fraction of an inch in terms of sixteenths of an inch. For example, 610 x 814 means that the can is nominally 6 10/16 inches in diameter and 8 14/16 inches high. In the case of a rectangular can measuring 610 x 505 x 411, the nominal dimensions of the end are 6 10/16 inches long by 5 5/16 inches wide, and the nominal height of the can is 4 11/16 inches.

MILITARY CUSTODIANS

Army - GL
Navy - SA
Air Force - 82

Civil agencies interest

GSA
VA

Preparing Activity

GSA-FSS

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